

Classification Assessment of Tumor Subtypes

26 March 2018

Updated: March 2026

In this group assignment, you will train a classifier that is able to predict subtypes of breast cancer tumors based on array CGH (aCGH) data. You will build a model in R or Python and make an estimate of the accuracy of its predictions on an independent test set. You need to explain your approach in a report and a presentation, and perform peer review on the draft report of one of the other groups.

Background

Breast cancer is a heterogeneous disease and classification of breast cancer tumors in their molecular subtypes has important implications for the treatment and prognosis. Three receptors play a pivotal role in these subtypes: the *Estrogen Receptor* (ER), *Progesterone Receptor* (PR) and *Human Epidermal growth factor Receptor 2* (HER2). After removal of the tumor, the pathology department of the hospital tests these samples for presence of ER, PR and HER2. They define the three main subtypes in breast cancer on which the treatment decision will be based are:

- HER2 positive: HER2+
- Hormone receptor positive (HR+): ER+ and/or PR+, and HER2-
- Triple negative (TN): ER-, PR- and HER2-

Each of the three subtypes reacts differently to different types of treatment.

Data

We provide you with a dataset with 100 breast cancer samples from the three subtypes. These samples are analyzed on a high-resolution array CGH platform with 244,000 probes per array that measures the quantity of chromosomal DNA. The pre-processing of the data has been done for you.

For each of these regions and each sample we give the call whether that region has a loss, normal, gain or amplification of the DNA: loss is -1, normal is 0, gain is 1, amplification is 2. Two files are provided, one containing the preprocessed aCGH data of the cancer samples (Train_call.txt) and the other containing the associated clinical outcome of these samples — the subtypes (Train_clinical.txt).

Building your classifier

Your task is to get a well-trained classifier for predicting the three breast cancer subtypes.

To achieve this, you might need some of the following steps.

1. Data purification, transformation, if necessary.
2. Feature selection, if necessary.
3. Choose machine learning methods (classifiers).
4. Train and validate the classifiers.

Programming and software

You are recommended (but not limited) to use R (3.3.2+) or Python (3.5+).

R

A number of machine learning R packages are available, such as CORElearn, RWeka and caret. You need to refer to the official documentation for usage.

If you are unfamiliar with R, you could start your work with the tutorial ([CATSRTutorial.pdf](#)) we have written for you, and we also advise you to use caret because we are able to provide some support for it.

You are required to save your trained classifier into a file by using 'saveRDS'. This model file (*.rds) together with R script needs to be submitted.

Python

If you opt to use Python, the Python library [scikit-learn](#) is recommended for the machine learning part. You may also need [NumPy](#) or [pandas](#) to transform the data; you can refer to the official documentation of these packages.

scikit-learn: [Quick start](#)

NumPy: [Quick start](#)

pandas: [10 minutes to pandas](#)

You are required to save your trained classifier into a file by using the '[joblib](#)' function of scikit-learn. This model file (*.pkl), together with the Python scripts, need to be submitted.

Classifier Assessment

To test your classifier's performance, you will be given another 57 samples (Validation_call.txt) for which you need to predict their subtypes (HER2+, TN or HR+). These samples will be shared before the draft submission. For the draft submission, you must provide predicted labels for these samples and an estimate for the number of correctly labeled samples. Shortly after we will publish the accuracy of your model on Canvas. Use the remaining time to improve your performance by retraining your model and refining your benchmarking scheme. Your final submission must include the updated labels and a re-estimated accuracy. Also, in **your final report you must give a detailed explanation of the changes you made to improve accuracy estimate** (and potentially model performance), and why you think this would improve it.

To make it clear here, both for the draft and final submission, you need to submit:

- 1) The predicted labels for each of the samples (file format given on [Canvas](#)) and
- 2) An estimate for the number of correctly labeled samples (out of 57).

The predictions file must meet the following criteria:

- File format: .txt
- Tab-separated values (TSV)
- Two columns: "Sample" "Subgroup"
- Include a header exactly as shown above (including quotation marks)
- Total number of lines: 58 (1 header + 57 predictions)

An example file is provided: Example_labels.txt

You can validate your file using:

```
python3 predictions_file_format_check.py <path_to_predictions.txt>
```

Schedule and key dates

Please see Canvas for an up-to-date schedule and hand-in dates for this assignment

Deliverables

- Report written in the style of a short paper [70%]
- Peer review of the draft report of another group (pass/fail)
- Predictions + accuracy estimate (+ R/Python-scripts, model file) [15%]:
 - o Format correct of predictions + model file [5%]
 - o Quality of predictions [5%]
 - o Accuracy estimate [5%]
 - o Full training and validation code - [pass/fail]

- Presentation [15%]

Submit your group assignment

This submission consists of two parts:

Part A

1. Draft of a report describing your research.
2. Peer review of another group's draft.
3. The final report.

Part B:

Predictions + accuracy estimate + scripts + model files.

Part A

A.1: Report draft is handed in through a Canvas assignment.

A.2: Peer review report is handed in through a Canvas assignment.

A.3: The final report is also handed in through a Canvas assignment.

Part B

This is also handed in through a Canvas assignment. However, as Part B consists of numerous files we require that you compress all your files into a .zip, .rar, or .tar file with file name '*groupXX*', e.g. group 5 should name their file 'group05.tar.gz'.

You should organize this compressed files into the following structure of subfolders and files:

- results (subfolder)
 - estimate.txt
 - prediction.txt
- model (subfolder)
 - model.rds or model.py
 - run_model.R or run_model.py
- code (subfolder)
 - ****.R* or ****.py*

'results', 'model' and 'code' are all subfolders.

'estimate.txt' is a text file only containing a number (between 0 and 57).

'prediction.txt' contains your sample predictions (file format given on [Canvas](#)).

'model.rds' is the file that saves your R model, 'model.pkl' is the one for Python.

'run_model.R' or 'run_model.py' is the script that reads the model file and outputs your prediction. You need to finish this script by filling in the template script (see [Canvas](#)).

When you finish the script, you need to run the fixed command line below to make sure the script works (file names in the command line can be different). You also need to check whether 'output.txt' has the same content as the 'prediction.txt' you will submit.

Rscript run_model.R -i unlabelled_samples.txt -m model.rds -o output.txt

OR

python3 run_model.py -i unlabelled_samples.txt -m model.pkl -o output.txt

The folder 'code' contains all the other scripts you have written for this project. Please keep them neat. They will be graded only as pass/fail, but should be able to reproduce your estimate.

Paper

For your (short) research paper, you need to choose a research question. We ask you to submit this question early on in the course, mainly to ensure that you are on the right track. Make sure to get feedback on this, before you write your draft. Please have a look back at the [writing lecture](#) from the course Fundamentals of Bioinformatics.

You should follow the Bioinformatics guidelines for writing an “original research paper”:
https://academic.oup.com/bioinformatics/pages/instructions_for_authors

Please find Word and LaTeX templates here:
https://academic.oup.com/bioinformatics/pages/submission_online

In addition, you should follow the outline described below:

Contents of paper (5-6 pages A4)

- Abstract (max 200 words), sections:
 - Motivation/background
 - Research questions
 - Results & impact
- Introduction
 - Set out the research question you are trying to answer. In grading, we will consider:
 - Is the research question clear?
 - Can it be answered with the given data and methodology used
 - Include references to other papers
 - Explain background of the data set (both experimental & preprocessing)
 - Explain the context (biomedical) of the research question
 - Explain if any similar bioinformatics approaches have previously been described
- Methods section
 - Explain and motivate which methods you have used to build your classifier
 - Explain how you have cross-validated your data. (In grading, we will consider if the cross-validation scheme makes sense)
- Results
 - Explain how well your classifier performs, based on your own benchmarking
 - Explain and justify the changes you made to improve the model's performance following the accuracy results in the draft submission.
 - You may compare multiple classifier approaches, or multiple feature selection methods.
 - If you were to mark a single region (biomarker) for classification, what would it be, and how well would you do?
- Discussion + conclusion (should be kept concise: no more than half a page)
 - Discuss any issues that may affect the results
 - Discuss the single best region you found (see above)
 - Clearly state what research question you have answered in this paper
 - Explain what impact your research has on future research
- Tables & Figures
 - Around 2-4 (in total) Figures and Tables
 - Make sure to explain all axes, labels, lines points, in the caption of your table / figure
 - Make sure to refer to each table / figure in the main text, and explain in the main text what can be seen from the figure
- Author contribution statement
 - Add what each group member did (see [PLOS ONE examples](#))

Grading of draft paper (5-6 pages A4)

Your draft will be peer reviewed by 4 other students. These will score you on the elements listed below, and give suggestions on how to improve the draft. These reviews will be combined by the teachers in a set of recommendations on what to improve for the final version of the paper.

- Context
- Research question
- Methods
- Contents/results
- Tables/figures

scoring: good / satisfactory / sufficient / insufficient

Grading of final paper (5-6 pages A4)

The final grade for the final paper will be based on the same rubric used for the thesis of the minor and major research project. To help you, we've outlined the content that we expect in each section above.

Presentation

For the presentation you should prepare exactly 4 slides per group:

1. cross validation/ testing scheme to estimate accuracy
2. (comparative) table of accuracy estimate(s) for submitted methods (and others if you wish)
3. methods (feature selection and classification) used for submitted prediction
4. rationale why you think the submitted method performed best

The presentation should last no longer than 8 minutes.